

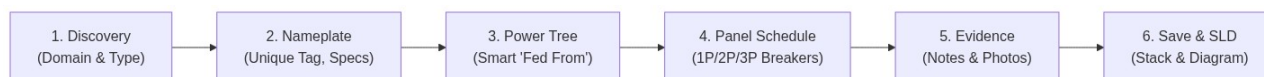
JEMBY JAMES Digital Twin - Field Data Collector Workflow & Field Guide

1. Executive Summary & Overview

The **JAMES Field Data Collector Workbench** is an interactive, responsive, multi-column survey platform designed for electrical engineers and field survey technicians operating on iPads and tablets during facility walk-through audits.

It bridges the gap between raw physical nameplates, panelboard breaker schedules, and photographic evidence, compiling them into a connected electrical Digital Twin backed by an atomic SQLite database (`model.db`) and a portable, Git-friendly JSON Lines companion file (`model.jsonl`).

2. Core 6-Step Field Survey Workflow



Step 1: Equipment Discovery & Archetype Selection

- **Column 1 (Domains):** Tap the primary equipment domain (*Power Sources, Transformers, Switches & ATS, Panelboards & MCC, Power Quality & UPS, Electrical Loads, Conductors, Metering, Renewables*).
- **Column 2 (Equipment Types):** Select the archetype (e.g. *Lighting Branch Panel LP, Main Distribution Panel MDP, Motor Control Center MCC, Dry-Type Step-Down Transformer XFMR*).
- **Auto-Incrementing Tags:** When selecting an archetype, JAMES automatically scans existing assets in the facility stack and assigns the next available unique suffix (e.g., if LP-1 exists, selecting Lighting Panel automatically defaults to LP-2).
- **Catalog Toggle:** Once familiar with the equipment in a room, tap **Hide Catalog** to collapse Columns 1 & 2 and grant 100% of the tablet display to the Nameplate Form and Facility Stack.

Step 2: Nameplate Specifications & Location

- **Unique Equipment Tag:** Every asset must have a unique tag (e.g. LP-1A, XFMR-1, ATS-1).
- **Duplicate Tag Guard:** Real-time validation checks for collisions across the facility stack. If a duplicate tag is entered, an amber **Tag in use** warning badge appears and the Save button is locked until a unique tag is provided.
- **Physical Room:** Electrical closet or room location (e.g. Level 1 - Electrical Room 101).
- **Electrical Physics:** Specify Voltage configuration (e.g. 480Y/277V 3Ø 4W, 208Y/120V 3Ø 4W), Main Bus Ampacity (e.g. 225A, 1200A), and Short Circuit Withstand (AIC / kAIC).
- **Automatic Conductor Sizing:** JAMES dynamically calculates NEC standard copper conductor sizes (e.g., 225A → 4/0 AWG Cu, 1200A → 4x 500 kcmil Cu).

Step 3: Upstream Hierarchy Link (Fed From)

- Select the upstream parent feeding asset from the **Fed From** dropdown.
- **Smart Physical Feeder Filtering:**
 - **Primary Sources** (Utility grid entrance, Standby generator, Solar PV) have no upstream feeder (-- None / Service Entrance --).
 - **Cycle Prevention:** Equipment cannot feed itself, nor can it be fed by its own downstream descendants.

- **Loads Never Feed Upstream:** Terminal loads (chillers, pumps, motors, EV chargers) are automatically excluded from all Fed From feeder lists.
- This deterministic link establishes the electrical tree:

UTIL-1 → ATS-1 → MDP-1 → T-1 → LP-1A

Step 4: Embedded Panelboard Breaker Schedule

- For panels, switchboards, and MCCs, Column 3 renders an interactive odd/even slot matrix (12, 18, 24, 30, 42, 54, 72, or 84 slots).
- Click any slot row to launch the **Circuit Breaker Inspector**:
- **1-Pole (120V / 277V L-N):** 1 slot on a single phase (A, B, or C).
- **2-Pole (208V / 480V L-L):** 2 vertical slots spanning alternating phases (e.g. Slots 1 & 3 = Phase A-B).
- **3-Pole (208V / 480V 3Ø):** 3 vertical slots spanning all 3 phases (Slots 1, 3, 5 = Phase A-B-C).
- **Trip Ratings & Types:** 15A to 225A ratings with MCCB, AFCI, GFCI, DF, MCP, or LSI trip units.
- **Smart Downstream Linking:** Link the breaker circuit to valid downstream loads (e.g. CH-1, PUMP-1). Upstream feeding ancestors and grid utility entrances are automatically excluded.
- **Spare / Space:** Click  Remove / Make Space to clear the breaker back to an unassigned spare slot.

Step 5: Field Notes, Multi-Photo Capture & Lightbox

-  **Snap Photo:** Directly invokes the tablet's hardware rear camera for instant nameplate or hazard photography.
-  **Add Photos:** Multi-select upload from the device image library.
- **Staging Tray:** Previews thumbnails with a red corner **X** button to discard unwanted shots before saving.
- **Fullscreen Lightbox:** Tap any thumbnail in a saved note to inspect high-resolution images, view timestamps, and download.
- **Zero Broken Links:** All photos are encoded directly into SQLite and the .jsonl companion file as self-contained Base64 Data URLs.

Step 6: Saving, Re-Editing, and SLD Generation

- Click **Save / Update** to commit the asset into **Column 4 (Facility Stack)**.
- **Unsaved Changes Safety Guard:** If you modify fields and accidentally switch assets, an interception modal provides [ Save & Switch], [ Discard & Switch], and [Keep Editing].
- **Facility Stack Grouping:** View by  **Room**,  **Power Tree**, or  **Equipment Type**.
-  **View SLD Diagram:** Compiles the Graphviz single-line drawing directly from the digital twin.
- **Archetype / Part #:** Selected from archetype templates or customized.
- **Electrical Ratings:** Real-time calculated fields (e.g. 3-phase FLA kVA calculations, Voltage system configurations 480Y/277V, 208Y/120V, etc.).
- **Upstream Connection:** Interactive dropdown selecting which parent panel or switchboard feeds this asset.

3. The 4-Column Panel Schedule Matrix

When a distribution panel, lighting panel, or switchboard is selected, the center column mounts an interactive **Breaker Schedule**:

```
+-----+
| POLE A/B/C | BREAKER (L) | TRIP (A) | [ L1 ] [ N ] [ L2 ] | TRIP (A) | BREAKER (R) | POLE A/B/C |
```

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Key Capabilities:

- **Odd / Even Phase Mapping:** Accurate NEMA panel layout reflecting alternating bus bar stabs:
- **Rows 1, 4, 7... → ■ Phase A**
- **Rows 2, 5, 8... → ■ Phase B**
- **Rows 3, 6, 9... → ● Phase C**
- **Multipole Gang Linking (1P, 2P, 3P):** Multi-pole breakers automatically lock adjacent bus stabs together across consecutive odd/even rows.
- **Real-Time Load Summaries:** Automatically sums Connected VA and computes total Panel Amperage load and phase imbalance metrics.
- **One-Click Phase Rotation Guide:**

Row	Left Slot	Phase	Right Slot	Typical Line Voltage
1	Slot 1	■ Phase A	Slot 2	120V / 277V to Neutral
2	Slot 3	■ Phase B	Slot 4	120V / 277V to Neutral
3	Slot 5	● Phase C	Slot 6	120V / 277V to Neutral
4	Slot 7	■ Phase A	Slot 8	120V / 277V to Neutral
5	Slot 9	■ Phase B	Slot 10	120V / 277V to Neutral
6	Slot 11	● Phase C	Slot 12	120V / 277V to Neutral

4. Inspection Checklists, Deficiency Notes & Field Sign-Off

Field technicians can perform standards-compliant physical and safety audits directly within the equipment drawer:

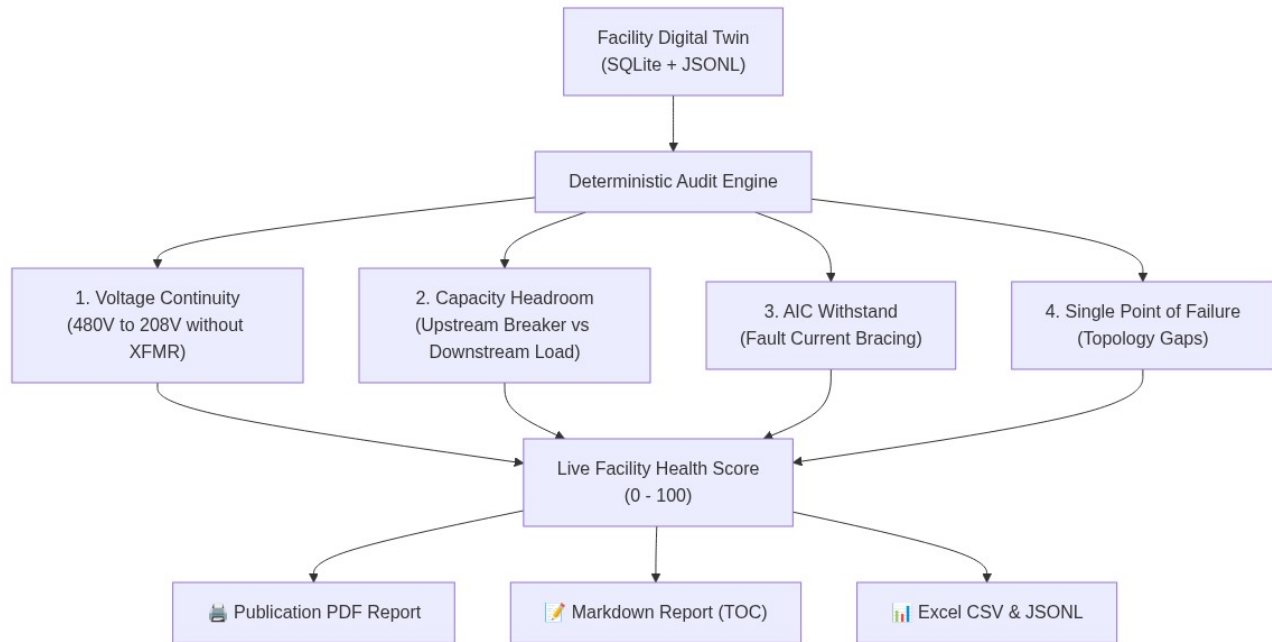
Key Features:

- **Domain & Archetype-Specific Protocols:** Checklists automatically load based on domain (e.g. *Panels, Transformers, Switches, Sources*) and equipment type (e.g. *Visual Condition, Working Space & NEC 110.26 Clearance, Arc Flash Labeling, Torque Verification, Grounding & Bonding*).
- **One-Tap Status Lifecycle:** Cycle items between **Pass (✓)**, **Deficient (■■)**, **N/A (—)**, and **Pending (■)**.
- **Auto-Revealed Deficiency Notes:** Marking any checklist item as **Deficient** immediately expands the notes textarea and triggers visual warning highlights.
- **One-Tap Preset Defect Chips:** Accelerate field note taking on tablets with pre-composed pithy deficiency tags:
 - + Label Missing
 - + Torque Unverified
 - + Clearance < 36"
 - + Missing Filler Plate
 - + Arc Flash Expired
 - + Torqued & Marked
 - + Clean & Vacuumed

- **Live Compliance Scoring:** Displays a real-time progress meter (e.g., 90% Compliant) across all completed and passed items.
 - **Inspector Sign-Off:** Record the certified field surveyor name, timestamp, and final compliance disposition.
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5. System Integrity Audit & Publication PDF Reports

The **System Integrity Audit** engine performs real-time engineering and code validation across the entire facility distribution network:



Key Capabilities:

- **Voltage Continuity Validation:** Detects unstepped voltage drops (e.g., a 480V panel feeding a 208V panel without an intermediate step-down transformer).
 - **Capacity & Headroom Analysis:** Verifies that upstream feeding breakers provide sufficient ampacity headroom for connected downstream equipment FLA / MCA ratings.
 - **AIC Rating Checks:** Flags devices with inadequate interrupting ratings relative to upstream available fault currents.
 - **Interactive Finding Inspection:** Click **Inspect Asset** ■ on any finding card to immediately open the equipment drawer for on-the-spot remediation.
 - **Multi-Format Reporting:**
 - ■■ **Publication-Ready PDF:** Server-side styled PDF with formal cover page, executive health scorecard, summary tables, detailed finding logs, and complete facility hierarchy.
 - ■ **Markdown with TOC:** GitHub-flavored markdown documentation with interactive Table of Contents.
 - ■ **Excel CSV:** Spreadsheet export for work-order tracking and external compliance records.
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6. Pro-Tips for Field Technicians

1. **Tag Integrity:** Duplicate equipment tags are blocked automatically; let archetype auto-incrementing speed up panel tagging (LP-1, LP-2, LP-3). 2. **Maximize Screen Real-Estate:** Use ■ Hide Domains on tablets to focus on the active breaker matrix and stack. 3. **Fast Guide Access:** Press ? on a physical keyboard or tap ■ Field Guide in the header. 4. **Emergency Discard:** Press Esc to instantly dismiss any open modal or photo lightbox. 5. **Lighting Conditions:** Switch between ■ Dark Mode (for dark electrical rooms) and ■■ Light Mode (for bright outdoor substations). 6. **One-Tap Deficiency Logging:** When marking an inspection item deficient, tap a preset chip (+ Label Missing, + Clearance < 36") to log issues in under 2 seconds.